

Review article

Modern scores for traditional tests - Review of the diagnostic potential of scores derived from word list learning tests in mild cognitive impairment and early Alzheimer's Disease

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ABSTRACT

Episodic memory impairment is one of the early hallmarks in Alzheimer's Disease. In the clinical diagnosis and research, episodic memory impairment is typically assessed using word lists that are repeatedly presented to and recalled by the participant across several trials. Until recently, total learning scores, which consist of the total number of words that are recalled by participants, were almost exclusively used for diagnostic purposes. The present review aims at summarizing evidence on additional scores derived from the learning trials which have recently been investigated more frequently regarding their diagnostic potential. These scores reflect item acquisition, error frequencies, strategy use, intertrial fluctuations, and recall consistency. Evidence was summarized regarding the effects of clinical status on these scores. Preclinical, mild cognitive impairment and mild Alzheimer's Disease stages were associated with a pattern of reduced item acquisition, more errors, less strategy use, and reduced access of items, indicating slowed and erroneous encoding. Practical implications and limitations of the present research will be discussed.

1. Introduction

Alzheimer's Disease (AD) is the leading cause of dementia, which already affects around 50 million people worldwide and will affect an estimated 150 million individuals by 2050 (Scheltens et al., 2021). One of the early hallmarks of AD are selective episodic memory impairments (Bäckman et al., 2005). Episodic memory impairments differ between healthy aging and early AD even when the two groups are matched for general cognitive performance (Cullum et al., 1995). Episodic memory is considered as one part of the long-term declarative memory system (Tulving, 1972) and comprises encoding (i.e., processing of new information), consolidation (i.e., transferring information into long-term memory), and retrieval (i.e., access and recall of items that are no longer in an active state) of personally experienced events (Tromp et al., 2015). The impairment of episodic memory in AD seems to originate mainly from deterioration in the processes of encoding and consolidation (Tromp et al., 2015; White and Ruske, 2002; Greene et al., 1996).

Preclinical and early stages of AD are often associated with amnesic mild cognitive impairment (aMCI) which is characteristic for the transitional stage between healthy ageing and dementia (Nordahl et al.,

2005), where focal deficits in episodic memory are already apparent (Collie and Maruff, 2000). These isolated impairments in episodic memory in AD and aMCI are associated with a characteristic early deterioration of the medial temporal lobe (MTL), which is related to memory functioning (Wolk et al., 2011; Nordahl et al., 2005). Other than in healthy aging where frontal areas are typically the first to degenerate, in AD the neurodegeneration starts in the MTL before spreading to cortical and frontal areas (Tromp et al., 2015). The isolated impairment of the episodic memory system already occurs in the pre-clinical stage, often several years before an AD diagnosis (Golby et al., 2005). This suggests that episodic memory functioning is a marker for beginning AD.

To date, several comprehensive reviews on cognitive and memory impairments of AD and aMCI exist in the literature with a focus on performance in word list learning tests (Albert, 1996; El Haj et al., 2016; Jahn, 2022; Tromp et al., 2015; Weitzner and Calamia, 2020). Previous reviews often compared and synthesized studies at the level of different test procedures rather than specific scores within one test paradigm (El Haj et al., 2016; Jahn, 2022). When synthesizing word list tests specifically, mainly total learning scores were investigated (e.g., Albert, 1996)

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and with rare exceptions also specific process scores like serial positioning scores (Weitzner and Calamia, 2020). The main aim of the present systematic review is to identify scores that were derived beyond total learning scores in word list learning tests and summarize evidence regarding their diagnostic usefulness in the MCI and early AD spectrum. Therefore, we will first introduce the learning paradigms and the different scores addressed in this review in the following.

1.1. Scores of memory performance in Word list learning paradigms

The word-list learning paradigm is widely adopted in AD research and clinical practice. Several word-list learning tests have been developed and became established tools, such as the Rey Auditory Verbal Learning Test (RAVLT; Rey, 1958), the California Verbal Learning Test (CVLT, Delis et al., 2000), the Hopkins Verbal Learning Test (HVL, Brandt, 1991), the CERAD word list (Morris et al., 1989), the RBANS word list (Randolph, 1998), the Philadelphia Verbal Learning Test (PVL; Bezdicek et al., 2014), or the International Shopping List Test (ISLT; Thompson et al., 2011). All these tests include a list of words (8–16 words), which is presented several times (learning trials) to allow the testee to gradually learn the words. Memory performance is measured with a subsequent immediate recall of each word list or with a delayed recall after around 10–30 min, depending on the specific test. However, these tests strongly vary in the length of word lists, the number of trials, permutation of word order (only in the CERAD word list), and the length of delay. Traditionally, memory performance is measured by focusing on total learning scores, e.g., the total number of recalled words from all words of the learning trials immediately after their presentation (immediate recall) and after a delayed time period (delayed recall) (Bruno et al., 2022). The immediate recall score represents encoding and learning of information (Vakil and Blachstein, 1993). The delayed recall score represents forgetfulness (Williams, 1968). While these scores and their diagnostic utility are well-established (Weissberger et al., 2017), process scores have been proposed in recent years and assessed regarding their diagnostic potential to detect early episodic memory decline. To keep the scope of this review within manageable limits, this review focuses on scores representing encoding and learning of new information, which are derived from learning trials. Scores derived from the delayed recall score are not considered in this review.

1.2. Scores derived from learning trials

Total learning scores in word-list paradigms represent a combined performance of selective sub-processes of episodic memory. Newly developed scores try to disentangle and quantify single memory processes, such as encoding and consolidation processes by looking at the exact retained list item, their position amongst other items, and their occurrence over the entire task. Scores that have recently been used and that aim at quantifying single memory processes in the learning trials include: (1) scores of item accumulation, including learning rates and slopes, which refer to the difference in the total number of items between two trials, and thus reflect how fast new items can be acquired (Dawidowicz et al., 2021); (2) scores of serial positioning effects, which analyze how many items are recalled from a specific position of the word list (Weitzner and Calamia, 2020); (3) intrusion errors, which reflect item confabulation (Barba et al., 2002), and (4) clustering scores which reflect strategies used to aid encoding (Hutchens et al., 2012; Wegesin et al., 2000), (5) gained and lost access scores, which capture the number of items that have been newly acquired or forgotten from one trial to another and thus allow to capture how efficient items are consolidated and maintained across several trials (Genon et al., 2013; López-Pérez et al., 2022). However, many different scores have been investigated, resulting in a large variety of different scores and varying computations that have not yet been systematically reviewed.

From a clinical perspective, a more precise characterization of

separate memory processes potentially increases diagnostic utility and supports early detection of AD-related cognitive impairment (e.g., Thomas et al., 2018). Since research in AD has shifted to the early phases of the disease, non-invasive diagnostic tools that achieve high diagnostic performance already in the early disease stages are required (Khoury and Ghossoub, 2019). Currently, there is a lack of systematically gathered information about the clinical utility of scores that quantify individual memory processes in learning trials in early AD. Therefore, the aim of this review is to identify scores that quantify individual memory processes in learning trials and summarize evidence on these scores to evaluate their diagnostic usefulness for characterization of episodic memory impairment in the early disease stages from healthy aging to mild AD. For this, scores that aim to quantify specific memory processes will be identified in the literature, selected, and grouped by the individual process they target, and finally summarized with regards to the effects of clinical status and disease progression on the specific scores.

2. Methodology

2.1. Search strategy

A systematic search in the databases “PubMed” and “PsycArticles” was conducted on August 19, 2022. The search terms included “serial positioning”, “verbal learning”, “mild cognitive impairment”, “AD” and “Aging” (full search terms can be seen in Table 1). No date or language restrictions were applied. Meta-analyses and reviews were excluded in the search in the PubMed database. Duplicates were removed. The systematic search was updated on December 11, 2023 to include recently published articles. The same search strategy as in the first search was used.

2.2. Screening

The screening was conducted in two steps: First, titles and abstracts were checked and all articles that clearly did not meet the inclusion and exclusion criteria (see section 2.4) were excluded. The selection was made conservatively and all articles for which it was not clear based on the abstract whether they could be used, were retained. Secondly, the full texts were viewed to check the eligibility of the given articles. All articles that did not meet the inclusion/exclusion criteria were removed

Table 1
Search strategy.

Search	Database	Search term	Number of results Aug 19, 2022	Number of results published between Aug 19, 2022, and Dec 11, 2023
#1	Pubmed	serial positioning AND (“AD” OR “mild cognitive impairment” OR “Aging”)	161	22
#3	Pubmed	(“Verbal learning” OR “word list learning”) AND (“mild cognitive impairment” OR “AD” or “Aging”)	2291	189
#2	PsycArticles	serial positioning AND (“AD” OR “mild cognitive impairment” OR “Aging”)	1	0
#4	PsycArticles	(“Verbal learning” OR “word list learning”) AND (“mild cognitive impairment” OR “AD” or “Aging”)	443	26
Total		= 3133	2896	237

from the analysis. No automation tools were used in the process.

2.3. Inclusion and exclusion criteria

2.3.1. Inclusion criteria

To be eligible for inclusion in this systematic review, studies had to fulfil the following criteria: 1) Assessment of verbal learning performance in an established and validated word list learning test (RAVLT, CVLT, CERAD, RBANS, Shopping List Test, HVLT, PVL, USC-REMT). The free and cued selective reminding test (Buschke, 1973) was not included because it relies on a different learning paradigm and is thus assumed to be not comparable with the other tests; 2) Inclusion of at least one cognitively impaired group (e.g., MCI or AD) or preclinical AD; 3) Scores that quantify specific memory processes (scores of item accumulation, serial positioning, intrusion errors, strategy use and intertrial fluctuations) are derived from the learning trials; and 4) Articles written in English or German. Studies that failed these criteria were rejected.

2.3.2. Exclusion criteria

From the initially included studies, studies were excluded if they 1) had no healthy or stable control group (e.g., a study was excluded if a cognitively impaired group (MCI or AD) was compared to other impaired groups such as Parkinson's Disease or Multiple Sclerosis), 2) included participants below the age of 45 in MCI or early AD groups, and 3) included participants that scored below 20 on the Mini Mental State Examination (MMSE; Folstein et al., 1983) OR higher than 1 on the Clinical Dementia Rating Scale (CDR; Morris, 1997) OR who were not classified as healthy, MCI or early AD using neuropsychological testing were excluded from the synthesis. Only the results for participants or participant groups that met the criteria are reported. Studies that only assessed groups that did not meet the criteria had to be removed from the analysis. Similarly, studies that only derived serial positioning scores had to be excluded. This exclusion criterion was added after the literature research was conducted. Initially, the goal was to see if a large component of the literature issued was also included in the Weitzner and Calamia (2020) paper. Since the overlap was substantial, it was decided that serial positioning is already covered and the scores were removed retrospectively, since no new insights can be generated beyond the existing review. However, serial positioning has proven to be of great diagnostic potential. Additionally, meta-analyses, reviews, animal, or veteran experiments as well as interventional studies and records without full text were excluded from the analysis.

2.4. Synthesis of results

A formal narrative synthesis was carried out. All studies were summarized by the score they assessed and grouped based on the verbal learning test that was used. Studies that analyze the same population were not generally excluded, if the studies analyzed different scores so that for each population and score the analysis was only considered once in the review. A study was excluded if it was conducted using the same dataset as another study and did not report any unique score that was not contained in previous analyses.

3. Results

A variety of different process scores mainly measuring the accumulation of items across trials, the frequency of intrusion errors, strategy use, and efficiency of item accumulation have emerged in research investigating diagnostic potential of word list tests. Different computations were used across studies.

3.1. Search results

The search identified 3133 individual studies across two databases

(see Fig. 1). After removal of duplicates, 2862 studies were eligible for screening. Of these, 2565 were initially excluded, leaving 297 studies that were assessed as full texts. From those, 261 were excluded, leaving 36 studies for the synthesis.

3.2. Changes in verbal episodic memory processes in MCI and early AD patients

To examine single memory processes, researchers have applied mainly scores reflecting learning, error performance, clustering, intertrial fluctuations and recall consistency. However, the computation of scores related to these processes is quite heterogeneous across studies and tests. Most of the studies have used scores of item accumulation. A tabular overview of the included studies per score can be found in the [Supplementary Material](#).

3.2.1. Scores of item accumulation

To capture learning processes, researchers analyzed the number of correctly recalled items across learning trials whereas mainly three types of scores were determined. There are no standardized terms for these three types of scores in the current literature. To compare the scores with each other, we use a standardized naming for the three types: learning rates, learning slopes, and trial-to-trial changes. We use “learning rates” to name scores that are calculated as a difference between the number of items in two trials. Typically, authors have used the difference of the lowest and the highest number of recalled items or the difference in the number of recalled items between the first and the last trial (e.g., Dawidowicz et al., 2021). Learning rates neglect information about the learning success from one trial to the next. In contrast, what we will refer to as “learning slopes” use this information by deriving parameters from an approximation of the learning curve mainly using regression-based methods, and hence providing one overall score across several trials (e.g., Gifford et al., 2015). Please note that due to inconsistent naming in the literature, some authors have referred to - what we call - “learning rates” as “learning slopes” and vice versa. We have renamed the scores to achieve a comprehensible synthesis. Trial-to-trial changes also use all information by averaging the number of correctly recalled items per trial and then determining significant increases or decreases between learning trials (e.g., Davis et al., 2013; Wang et al., 2013). In three articles, the exact computation of the item accumulation scores used was not provided (e.g., Bondi et al., 1995; Loring et al., 2016; Warren et al., 2021).

Learning rate. A substantial number of studies investigated the learning rates in early AD and MCI populations (Chen et al., 2022; Ciafone et al., 2022; Dawidowicz et al., 2021; Estévez-González et al., 2003; Foster et al., 2009; Genon et al., 2013; Gifford et al., 2015; Hammers et al., 2021; Incalzi et al., 1995; Ter Huurne et al., 2023; Thomas et al., 2018; Wang et al., 2013). Studies reported differences in learning rates depending on the disease progress (Ciafone et al., 2022; Dawidowicz et al., 2021; Estévez-González et al., 2003; Gifford et al., 2015; Hammers et al., 2021; Incalzi et al., 1995; Wang et al., 2013). Learning rates (difference between the first and last trial) are significantly reduced in MCI in comparison to HC groups (Estévez-González et al., 2003; Ciafone et al., 2022; Gifford et al., 2015; Wang et al., 2013). The same is true for participants who were deteriorating in their cognition in comparison to participants with stable cognitive functions (Estévez-González et al., 2003; Dawidowicz et al., 2021). However, it is important to note that in the study by Dawidowicz et al. (2021) participants who later declined were on average 12 years older than the cognitively impaired sample that did not decline. Diagnostic potential of learning rates was investigated using an classification model which also included further process scores and scores from a semantic verbal fluency test (Ter Huurne et al., 2023). The model achieved an AUC of 0.89 in classifying MCI and SCI. The specific contribution of learning rates to the model is unclear. The extent to which the learning rate has diagnostic performance beyond other clinical measures used in the field

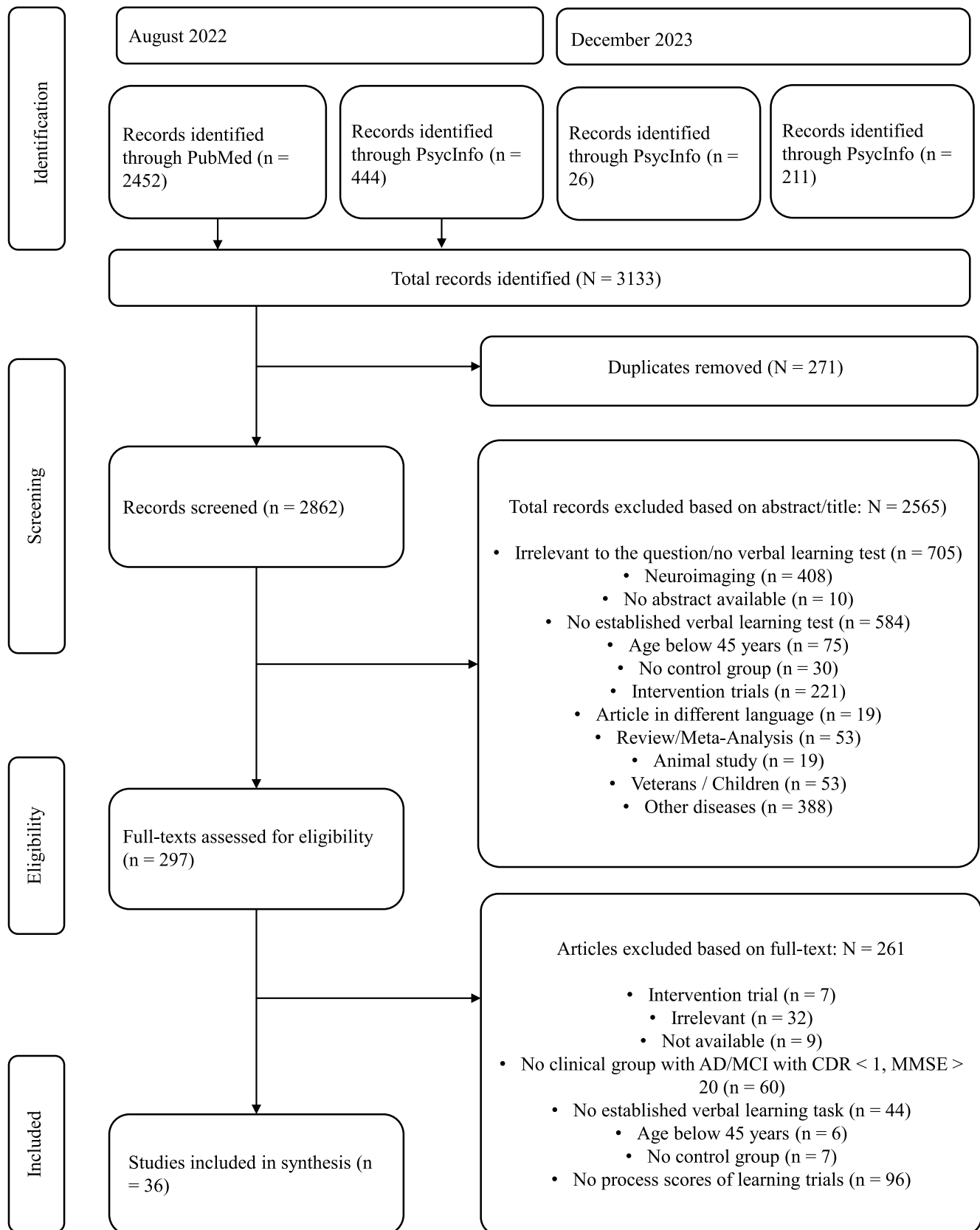


Fig. 1. PRISMA Flowchart for article selection.

is questionable. A study by Thomas et al. (2018) shows that the learning rate does not significantly contribute to the prediction of MCI conversion, when APOE status and total learning scores are already included in the model.

Some studies varied the number of learning trials used to calculate the learning rate (e.g., Chen et al., 2022; Wang et al., 2013). When reducing the number of trials to three, only MCI participants with lower overall cognitive scores performed worse than HCs (Chen et al., 2022; Wang et al., 2013). In both studies, participants were very similar in their mean age, and in both studies, they were defined as MCI based on an amnesic impairment (Chen et al., 2022; Wang et al., 2013). The main difference between studies lies in the level of cognitive impairment: MCI participants in Wang et al. (2013) had slightly lower MMSE scores than Chen et al.'s participants ($M = 25.73$ vs. $M = 27.37$). The shortening of the learning rate to only the first three trials seems to be sensitive to amnesic cognitive impairment only at lower overall cognitive scores. In line with this, Wang et al. (2013) also reported a higher discriminative power for learning trials two and three in comparison to the first one. Hence, the diagnostic potential of the learning rate seems to be related to the number of learning trials with increasing discriminative power with more learning trials.

Finally, some authors used adapted versions of the learning rate. Incalzi et al. (1995) took the difference of the pooled first and last two trials and reported significantly reduced scores for mild AD patients in comparison to HCs. Hammers et al. (2021) calculated two different item accumulation scores on the HVLt and the RBANS: the learning rate and the learning ratio, which takes the number of items learned in Trial 1 and divides them by the number of items yet to be learned after Trial 1. This score diminishes the influence of the initial learning performance (Trial 1) on the calculation of the actual increase in items. Since, in the commonly used learning rate calculation (Trial 5 - Trial 1), the performance in the first trial determines how much more can be gained, these scores may be a distorted representation of item accumulation. The learning ratio was significantly lower in HC compared to MCI and AD groups in the three-trial long HVLt and the four-trial long RBANS. The learning rate in the HVLt was significantly reduced in AD participants when compared with HCs. No differences emerged between HC and MCI and between MCI and AD groups, which means that the learning ratio was more sensitive to the different clinical states in comparison to the learning rate in the HVLt. In this study, the learning rates calculated on the RBANS were somewhat more sensitive to mild cognitive impairments than learning rates calculated on the HVLt. The learning ratio was slightly more sensitive to MCI on the HVLt than the learning rate. The RBANS has one more trial than the HVLt. In summary, the learning rate works better than the learning ratio in a longer test and the learning ratio works better than the learning rate in a shorter test. Taking out the influence of the first trial, potentially increases diagnostic potential of learning rate scores in shorter tests. In the opposite approach, the learning efficiency index, which puts more weight on the first trial, reveals lower scores in participants with cognitive impairment in comparison to HCs in the five-trial long CVLT (Genon et al., 2013).

In summary, different versions of the learning rate are sufficiently sensitive in differentiating between clinical groups. Overall, learning rates over a larger number of trials (more than three) seem to be more sensitive than learning rates considering a lower number of trials. Taking the influence of the first trial out of the computation may increase diagnostic performance in the shorter tests.

Learning slopes. Slope scores including regression-based slopes, early and late slopes (calculated on the early and late trials), and the peak slope, which is derived based on the first and the learning trial with the most remembered items, were only rarely implemented.

Using regression-based approaches, cognitively impaired groups had lower slope scores than HCs (Estévez-González et al., 2003). For the peak slope, mixed results were found (Gifford et al., 2015; Foster et al., 2009). Methodologically, these studies are hard to compare. Gifford et al. (2015) used the RAVLT, which at 5 learning trials is significantly

longer than the HVLt used by Foster et al. (2009). Furthermore, Gifford et al.'s sample is more than 10 times larger than that of Foster et al., which might have contributed to the significance of the effect. In the same study, MCI participants had significantly lower early and late slope scores than the HC group.

Trial-to-trial changes. It has been shown that single trial scores do not differ in the preclinical stage (Chen et al., 2002; Wehling et al., 2007). With the onset of cognitive impairment, significantly lower increases in the number of correctly recalled words from trial to trial can be observed in comparison to HCs (Campos-Magdaleno et al., 2016; Chen et al., 2011; Fernández-Ballesteros et al., 2007; Greenaway et al., 2006; Klekociuk and Summers, 2014; Mitrushina et al., 1994a; Ribeiro et al., 2007).

Researchers have included further scores that approximate increments of items across trials. A Bayesian model identified two variables that approximated memory processes of the encoding and transfer of items to long-term memory storages and differed between stable HCs and HCs that later declined to MCI or AD (Bock et al., 2021).

In sum, researchers have implemented different scores that assess the accumulation of items across learning trials. Among these, learning rates, slope scores and trial-to-trial changes are frequently implemented. However, all scores of item accumulation were significantly affected by clinical status. Especially learning rates and slope scores proved high sensitivity for cognitive impairment even at an early stage with pre-clinical and MCI individuals showing reduced item accumulation in comparison to HCs.

3.2.2. Error frequencies

Error scores that were applied in the literature are intrusions (words that are mentioned although they were not on the list) and perseverations (a correct word is named more than once). Although words that are forgotten could be counted as an error, these scores are grouped and discussed together with the total learning scores and intertrial fluctuations and are not mentioned here.

Perseverations. Two studies were identified that assessed the number of perseverations in cognitively impaired samples (Bennett et al., 2006; Chen et al., 2011). Bennett et al. (2006) found no significant difference between aMCI and HC groups. In contrast, Chen et al. (2011) reported some significant differences in the number of perseverations between MCI and HC groups depending on the education status and age: MCI participants with higher education and younger than 70 years, showed more perseveration errors than HCs (Chen et al., 2011). However, for participants that have received lower education, HCs did more perseveration errors than the MCI group. A substantial difference between studies is the inclusion of MCI (Bennett et al., 2006) vs. aMCI participants (Chen et al., 2011). While an exclusive amnesic impairment does not affect the frequency of perseverations (Bennett et al., 2006), mild cognitive impairment, including multiple phenotypes, such as impairments in executive functions, can be - under consideration of demographic covariates - associated with higher perseveration errors (Chen et al., 2011). Hence, other cognitive impairments, such as declining planning, inhibition and structuring of the verbal output might have caused higher perseveration errors than amnesic memory problems.

Intrusions. Analysis of the literature reveals that MCI groups make more intrusion errors than HCs and that mild AD groups make more intrusion errors than HCs and MCI groups (Campos-Magdaleno et al., 2016; Chen et al., 2011; Greenaway et al., 2006; Mizrahi et al., 2023; Ribeiro et al., 2007; Wehling et al., 2007). Additionally, more intrusion errors were found when individuals who decline at later test time points were compared with individuals who remain cognitively healthy (Thomas et al., 2018). Concerning the quality of the cognitive impairment, especially aMCI seems to be related with increased intrusion errors (Campos-Magdaleno et al., 2016). Taking the percentage of intrusions instead of the total number of intrusions led to the same results (Greenaway et al., 2006). Studies that included a smaller sample

size or participants with low levels of cognitive impairment failed to find a difference in the amount of intrusion errors between clinical groups (Bennett et al., 2006; Lekeu et al., 2010; Mitrushina et al., 1994b). Mizrahi et al. (2023) compared English and Spanish speaking participant groups. In the English-speaking group, MCI participants made significantly more intrusion errors than HCs, which was not found in the Spanish-speaking group. It is important to note that the English-speaking group completed a longer test battery, which likely led to more fatigue which might have impacted the performance. Additionally, the CERAD was translated to Spanish, which resulted in words that were on average longer than the English words, so that the two groups are probably not comparable. However, it is likely that language affects the results.

In sum, cognitive impairment seems to lead to increased error frequencies. Especially amnesic impairments are associated with a higher intrusion frequency. Contrary, amnesic impairments do not seem to lead to higher perseverations.

3.2.3. Clustering scores

Clustering scores are based on the order in which words are recalled. It is assumed that the order is not arbitrary but results from an attempt of the individual to structure the presented material. The specific order allows tracing back to a specific cognitive strategy that the individual uses to structure the items (Blachstein and Vakil, 2021). The three commonly recognized strategies are semantic, serial, and subjective clustering. Semantic clustering means that items are structured based on semantic proximity or commonality, such as an overall category (Ribeiro et al., 2007). Serial clustering means that items are structured following the order of the presented list (Price et al., 2010). For example, if the original list would be “mice - drum - house” and a participant would recall “house - mice - drum” this would be a serial cluster with two words (mice, drum). Subjective clustering typically refers to all clusters of words that are recalled together across subsequent trials that do not follow any objective clustering approach (Bennett et al., 2006). Regarding strategy use, the comparison of individual tests is required since only the CVLT and HVLIT provide a semantic structure of the word list, while serial and subjective clustering can be used in all word list learning tests. This can influence the results, e.g., higher serial clustering is observed in the RAVLT than in other tests (Johnson, 2012).

Semantic clustering. In the selected studies, semantic clustering has only been computed on the CVLT and HVLIT, in which items are already semantically related as they are part of four different categories. Regarding the specific computation of clustering scores, raw scores, and chance-adjusted scores can be differentiated. Raw scores represent the uncorrected, total number of clusters as defined by at least two words from the same semantic category in adjacent positions (Zhang et al., 2019). Chance-adjusted scores are raw scores corrected for the number of clusters that would be expected by chance. Mainly two different corrections have emerged: (1) The “recall-based clustering” which uses the words recalled by the participant as a reference to correct the raw score, and (2) the “list-based clustering” which uses the words from the presented list (McLaughlin et al., 2014). While the recall-based clustering assumes that the structuring happens after the words are retrieved from memory, the list-based clustering assumes that the structuring happens during the learning process (McLaughlin et al., 2014).

In general, semantic clustering improves memory performance and leads to a higher number of total recalled items in all groups (Price et al., 2010; Zhang et al., 2019). Age was commonly controlled for in the analysis of effects of cognitive impairment.

For the comparison of ApoE4-positive and ApoE4-negative participants, inconsistent results were reported (Bondi et al., 1995; Chen et al., 2002). Inconsistent results could be due to the different computations that were used (semantic clustering ratio vs. chance-adjusted scores) or due to a small effect in this preclinical stage.

In symptomatic stages, a reduction in using a semantic clustering strategy becomes evident (Campos-Magdaleno et al., 2016; Ribeiro

et al., 2007; Hutchens et al., 2012; McLaughlin et al., 2014; Lekeu et al., 2010; Price et al., 2010; Zhang et al., 2019). MCI individuals show a reduced increase in clustering across trials so that MCI individuals have significantly lower scores in the later trials (McLaughlin et al., 2014). Reduced semantic clustering is consistently found across different computations including list-based and recall-based corrections, demonstrating that the structuring of verbal material seems to be impaired at the learning and at the recall stage, which indicates that the impairment of the structuring ability is not bound to a specific memory stage.

Serial clustering. For serial clustering, unidirectional and bi-directional computations exist. Unidirectional computations only count two words as a cluster if they have the exact order as in the presented list. Bi-directional computations consider word clusters if the order is inverted. For example, if the list is “mice - drum - house” and the participant recalls “drum - mice - house” this would be a serial cluster of two words (drum, mice).

Studies consistently reported no difference in serial clustering between clinical groups either in the symptomatic or preclinical stage and healthy controls (Chen et al., 2002; Lekeu et al., 2010; Price et al., 2010; Zhang et al., 2019). In the same studies, semantic clustering scores revealed differences between clinical groups ranging from MCI to mild dementia, and had overall higher scores than serial clustering, indicating that all groups engaged more in semantic clustering. The higher use of semantic clustering was associated with the lower use of serial clustering scores in all groups, making it difficult to meaningfully differentiate between the groups. In tests that provide a semantic structure, like the CVLT or the HVLIT, semantic clustering seems to be the more popular strategy. Hence, it is difficult to interpret the results on serial clustering in these tests. Another study on serial clustering that was obtained from tasks that do not provide a semantic structure also included further scores and only reported overall classification performance values (Ter Huurne et al., 2023).

Subjective clustering. For subjective clustering, unidirectional and bi-directional computations exist. Unidirectional computations only count two words as a cluster if they are named in the same order across two subsequent trials. Bi-directional computations also consider word clusters if the order has changed over two consecutive trials. If an individual says “hat - turkey” in trial n and “turkey - hat” in trial $n + 1$ only a bidirectional score would count the answer as a cluster. Subjective clustering scores can then again be corrected for the number of items in a list or for the number of items that are correctly recalled by the individual.

For subjective clustering, the literature is sparse and inconclusive. One study found no difference between a small sample of amnesic MCI participants and HCs in recall-adjusted bidirectional clustering scores (Bennett et al., 2006), whereas the other study found less bidirectional subjective clustering for amnesic MCI participants than for HCs (Hutchens et al., 2012).

In addition to semantic, serial, and subjective clustering, researchers have recently used a temporal clustering approach, which considers items that are recalled close together in time as a temporal cluster (Dörr et al., 2023). The study indicates that scores that measure the time between individual words and between temporal clusters could differ between executive and amnesic MCI. More research is needed to understand the effect of amnesic MCI on temporal clustering. It might also be useful to map temporal information with other clustering analyses to understand where the temporal structure comes from.

In summary, clustering scores are rather inconsistently related to clinical status. Substantial differences emerged between cluster types and the effect of cognitive impairment. Reduced semantic clustering was observed with increased cognitive impairment, independent of the corrections that were applied. In contrast, serial and subjective clustering scores did not differ between HCs and cognitively impaired participants. Studies that assess multiple cluster types reveal that semantic clustering is used more frequently than other cluster types when the test

provides a semantic structure. In addition, serial clustering is generally less used than semantic or subjective clustering strategies resulting in generally lower scores in all groups, and by this making it difficult to meaningfully differentiate between the groups.

3.2.4. Scores of consolidation efficiency

Researchers have applied scores capturing intertrial fluctuations and consistency of recall. Intertrial fluctuations typically calculate on a trial-to-trial basis how many items were newly acquired and how many items were forgotten although they were correctly recalled in a previous trial. Intertrial fluctuations thus approximate how efficient the learning and consolidation process is in the short delay from one trial to another (Genon et al., 2013). “Gained access” scores are the number of words that are recalled on trial $n+1$ that were not recalled on trial n divided by the number of items that were not recalled on trial n , summed across all trials. Similarly, “lost access” is the number of words that were not recalled in trial $n+1$ although they were recalled at trial n divided by the number of items recalled on trial n , summed across all trials (see Fig. 2). Sometimes these two scores are referred to as “additions” (gained access) or “omissions” (lost access) (e.g., Blachstein and Vakil, 2015).

While intertrial fluctuations look at the trials and investigate whether items were gained or lost, scores of recall consistency look at a specific item and calculate how often the item is recalled in subsequent trials.

Intertrial fluctuations. Individuals with prodromal AD including amnesic MCI seem to show a reduction in gained access scores in comparison to HCs (Genon et al., 2013; López-Pérez et al., 2022; Wang et al., 2013). In contrast, Sugarman et al. (2014) reported higher gained access scores for participants who declined more than one Standard Deviation from baseline in either the Dementia Rating Scale or the RAVLT total learning scores. In comparison to other studies, this definition of cognitive decline is rather liberal and may not mark a clinically significant decline. The higher gained access scores may thus represent an effect of ageing instead of a sign of MCI or beginning AD.

For lost access scores, the literature is inconsistent. Genon et al. (2013) and López-Pérez et al. (2022) found no significant difference between MCI and HC groups. However, Wang et al. (2013) reported significantly higher lost access scores for aMCI in comparison to HCs. While the assessed populations are comparable in terms of age and overall cognition, different tests were used: Wang et al. (2013) used the RAVLT which uses semantically unrelated common nouns and Genon et al. (2013) and López-Pérez et al. (2022) used the CVLT, which contains semantically related words. It is possible that a semantic structure influences intertrial fluctuations, for instance that semantic categories, which can work as cues prevent high lost access scores even in impaired populations.

In sum, amnesic cognitive impairment seems to have reduced gained access scores, indicative of impaired acquisition of novel information. Evidence is less clear for lost access scores. More research might

be needed on intertrial fluctuations, also addressing the preclinical disease stage.

Recall consistency. Further measures of recall consistency were applied infrequently. A percentage score of items that were recalled in two subsequent trials showed no difference between participants with and without a genetic predisposition for AD (Chen et al., 2002). Ter Huurne et al. (2023) included a constancy learning index together with other process scores from verbal learning and a verbal fluency task in a machine learning classifier and differentiated successfully between MCI and SCI participants. The specific computation and contribution of the score is unclear. In comparison to intertrial fluctuations, recall consistency has rarely been implemented and little is known about the effect of clinical status on recall consistency.

4. Discussion and Future Directions

The aim of the present review was to identify scores that capture single memory processes in word list learning tests and to determine their ability to differentiate different stages from healthy aging to mild dementia. Scores derived from the learning trials that occur particularly frequently in the literature are 1) scores of item acquisition, including learning rates, learning slopes, and trial-to-trial changes, 2) scores of error frequencies, including intrusion and perseveration errors, 3) scores of clustering, which refer to the structuring of the presented material and primarily include semantic, serial, and subjective clustering, 4) scores of consolidation efficiency, which include intertrial fluctuations and recall consistency. The effects of clinical status were summarized for each score (see Table 2). In the following, the diagnostic potential of these scores will be summarized and discussed.

Symptomatic disease stages, including mild cognitive impairment and dementia stages are characterized by reduced trial-to-trial increases, reduced learning slopes and less semantic clustering and more intrusion errors. Very little evidence is reported for the preclinical stages, defined by a genetic predisposition or positive biomarker results in the absence of cognitive symptoms (Dubois et al., 2016). Future research should also seek to recruit participants from the pre-clinical stages. Since the early administration of medication is most effective (Porsteinsson et al., 2021), diagnostic options are needed that give an indication of a potential disease at an early stage. In addition to invasive methods that are impractical to be used large-scale, process scores derived from verbal learning and memory tests could provide a low-burden alternative.

When looking at individual scores, the increase in items across trials as measured with learning rate, slopes, and trial-to-trial changes is significantly reduced in cognitively impaired persons in comparison to HCs starting relatively early in the disease progression (Bondi et al., 1995; Dawidowicz et al., 2021; Estévez-González et al., 2003; Gifford et al., 2015; Incalzi et al., 1995; Wang et al., 2013). In the current literature, various scores and calculations have been used for the accumulation of items across learning trials. Also, there is no standardized

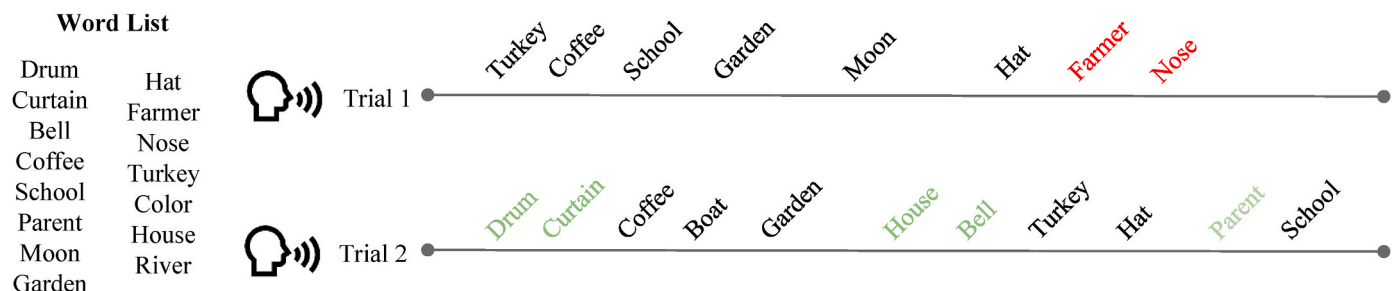


Fig. 2. Example of the calculation of gained and lost access scores

Note. Words printed in red are items, which were lost from trial 1 to trial 2. In this example, two words out of eight that were recalled on trial 1, were lost. The lost access score for this trial would be $\frac{2}{8} = 0.25$. Words printed in green are items which were gained from trial 1 to trial 2. In this example, five words were gained. The number of items that were not recalled on Trial 1, results in the number of items in the word list minus the number of items that were recalled on Trial 1 and is $15 - 8 = 7$. The gained access score for this Trial would be $\frac{5}{7} = 0.71$.

Table 2

Overview of results.

Score group	Score	HC vs. preclinical AD	HC vs. cognitively impaired (CI)	
		Summary of results	Summary of results	Direction of results
Item acquisition	Learning rate	/	+	HC > CI
	Learning slope	/	+	HC > CI
	Trial-to-trial changes	-	+	HC < CI
Error frequencies	Intrusion errors	/	+	HC < CI
	Perseveration errors	-	+	HC < CI
Clustering scores	Semantic clustering	~	+	HC > CI
	Serial clustering	-	-	
	Subjective clustering	/	~	
Consolidation efficiency	Gained access	/	+	HC > CI
	Lost access	/	~	
	Recall consistency	-	/	

Note. + majority of studies report significant results, /not assessed, - majority of studies report insignificant results, ~ inconsistent findings.

naming of learning rate and learning slope (terms that have been used in a standardized way here). To ensure that the subject area can be better understood, and aggregated, specific descriptions of the computations should be provided and standardized names and computations should be used in the literature. Despite the variation in naming and computations a vast majority of studies found differences in item accumulation scores between cognitively impaired and healthy individuals, indicating that AD-related cognitive impairments affect the ability to accumulate knowledge. Learning in word list learning tests seems to be associated with a broad neural network, including the cingulate and frontal cortex as well as the temporal lobe. Especially the hippocampus, which declines in AD pathology, shows a substantial correlation with learning in the RAVLT (Chang et al., 2010). The involvement of further areas may also indicate that not only the memory system, but executive and working memory abilities are needed during word list learning.

Perseveration errors seem to be preserved in aMCI but increased in non-amnesic MCI subtypes (Bennett et al., 2006; Chen et al., 2011), indicating that recalling words that were already recalled is more associated with a planning and strategy problem, than a pure memory impairment. In contrast, a higher number of intrusion errors was observed in MCI, especially in amnesic MCI (Campos-Magdaleno et al., 2016; Chen et al., 2011; Greenaway et al., 2006; Ribeiro et al., 2007; Wehling et al., 2007). Intrusion errors can be interpreted as an insufficient separation of distinctive memory traces - also referred to as pattern separation (Gallagher and Koh, 2011). Pattern separation is associated with structures known to decline in AD, including the entorhinal cortex, dentate gyrus, and the CA3 region of the hippocampus. In line with this, amnesic MCI participants experience a progressive worsening of pattern separation in comparison to healthy aging controls (Gallagher and Koh, 2011). Hence, the high number of intrusions in participants with amnesic MCI may capture clinically relevant decreases in pattern separation.

Further scores assessing the order in which items are recalled, demonstrate that the strategy to structure the presented list differs between healthy and cognitively impaired individuals. Cognitively impaired individuals engage less in semantic clustering than HCs in word list tests that contain semantically related words (Campos-Magdaleno et al., 2016; Ribeiro et al., 2007; Hutchens et al., 2012; McLaughlin et al., 2014; Lekeu et al., 2010; Price et al., 2010; Zhang et al., 2019). However, differences were only found for symptomatic

stages. Next to semantic clustering, subjective and serial cluster scores were examined in the literature. While a reduction was reported for subjective clustering (Bennett et al., 2006), serial clustering was not affected by impairments (Price et al., 2010; Zhang et al., 2019; Lekeu et al., 2010). Studies that assess more than one clustering strategy revealed that the serial clustering strategy is used less than subjective or semantic clustering in general by all age groups and disease stages (Blachstein and Vakil, 2021; Price et al., 2010). Since the scores are generally low in all groups, comparisons between groups did not reveal any significant differences indicating that serial clustering scores are the least used clustering strategy which impacts their diagnostic utility.

A distinctive pattern of gained access scores occurs in cognitively impaired individuals, indicative of a disease-related reduction in consolidation efficiency (Genon et al., 2013; López-Pérez et al., 2022; Wang et al., 2013). Participants with amnesic cognitive impairment in the MCI or mild AD stages show lowered gained access scores. This can be interpreted as an impairment of the initial encoding of new information and accumulation of knowledge (Wang et al., 2013; Genon et al., 2013; López-Pérez et al., 2022). These results thereby demonstrate that item-level analysis can increase the amount of useful information that can be obtained when analyzing word list learning tasks. However, evidence is less clear for lost access scores.

In summary, clinical disease-related changes are evident in scores of item acquisition, error frequencies, and semantic clustering. A reduction in gained access scores also indicates difficulties with the acquisition of knowledge instead of a consolidation problem.

4.1. Limitations

Most studies did not provide imaging or biomarker information or used them to diagnose participants. Therefore, results cannot be linked specifically to AD and its underlying neurological and pathological states. Additionally, many studies investigated MCI individuals. Although MCI is often a precursor of AD and 70–80% of MCI individuals have an underlying AD pathology, other indications cause MCI as well (Petersen, 2009). Without imaging or biomarker information, results on MCI cannot be clearly associated with AD pathology.

Secondly, the present work synthesizes evidence of a rather heterogeneous study population. Heterogeneity arises from 1) variety in pathology, since MCI can originate from multiple pathologies and 2) variety in the considered demographic variables. With an age range of 45–97 years, individuals with different lifestyles and most likely with different overall cognitive abilities are compared in this analysis. Since some studies already indicate an association of the scores reported here with age and education (e.g., Chen et al., 2011; Hammers et al., 2022; Lundervold et al., 2014), it is likely that some effects that are reported here are impacted by factors other than just the clinical status. Although we have considered age as an influential factor in the synthesis, it was not the focus of this work and no information about the effect of age and its interaction with the presented scores was systematically gathered. Hence, future reviews or meta-analyses could also look at the influence of age and other demographic characteristics on the relationships between process scores and cognitive impairments.

Thirdly, future synthesis could add scores that were derived from the delayed recall trial to the scores discussed here. Since the delayed recall trial examines a different ability (remembering previously learnt words), it might be interesting to investigate which scores are particularly suitable for assessing impairment in pathological aging. Building on this, it could also be investigated which scores of delayed recall trials add something to diagnostic decisions beyond the process scores discussed here.

Cultural factors were not considered in this analysis. Although no effect of cultural background was found for many learning scores in two different ethnicities (Lim et al., 2012), the cultural independence of the scores is not confirmed and not applicable to further cultures. Moreover, since the cultural background was not controlled, the samples that were

compared for this analysis are rather heterogeneous. Heterogeneity in the samples might have contributed to inconsistent and diverging results. Additionally, as indicated by Mizrahi et al. (2023), language might affect the findings discussed here leading to reduced comparability of studies. It is important to investigate the robustness of specific tests, administrations and individual scores against the influence of languages to receive an estimation of generalizability of results.

A varying number of studies is published for the individual features with healthy elderly and clinical populations with some features being assessed infrequently. Especially, results in the preclinical stage are based on a comparably lower number of studies. To be able to reliably estimate the effect size of group differences of individual scores, a meta-analytical aggregation would be needed. For this, a larger number of more homogenous studies is required for most of the scores, including clustering, learning slopes, and recall consistencies. Additionally, a meta-analytical aggregation would also require more consistent computations of scores. Currently, there is no consensus in terms of computations and names in the scores derived from learning trials making a synthesis more difficult. In future research standardized names for individual scores should be established.

4.2. Practical implications and conclusion

The results of this review have practical implications as they reveal that scores obtained from word list learning tests have diagnostic potential for cognitive impairment and AD even in the early disease stages. All scores that were analyzed were derived from the learning trials and already show differences between healthy individuals and a clinical population. Hence, a certain diagnostic performance seems to be given if only the learning trials are administered, which might allow for the omission of the delayed recall trial, which would significantly reduce test duration and would thus allow for a more time-efficient and low-burden assessment. Additionally, a substantial number of studies investigated the change of individuals in a longitudinal design with promising results (e.g., Estévez-Gonzalez et al., 2003; Genon et al., 2013; Lekeu et al., 2010), indicating that the individual decline and a longitudinal follow-up can provide large amounts of information about the cognitive state of an individual and thus hold large diagnostic potential.

To conclude, the evidence on the scores collected here demonstrates that scores of item accumulation, intrusion frequencies, consolidation efficiency, and strategy use can provide useful information about episodic memory processes in the early AD spectrum. The clinical population from preclinical to mild AD stages displayed a pattern of reduced acquisition of items, more intrusions, less semantic clustering and reduced gained access, indicating slower and erroneous encoding.

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CRediT authorship contribution statement

Simona Schäfer: Writing – original draft, Writing – review & editing, Methodology, Investigation, Data curation, Conceptualization. **Johannes Tröger:** Writing – review & editing, Conceptualization. **Jutta Kray:** Writing – original draft, Writing – review & editing, Supervision.

Declaration of Competing interest

SS and JT are employed at the digital biomarker company ki:elements GmbH. JT owns shares of the company. JK has no conflicts of interest.

Data availability

No data was used for the research described in the article.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.neuropsychologia.2024.108908>.

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